

An incremental elastic-plastic cohesive constitutive model with considering damage

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The interface has a significant influence on the macroscopic mechanical behaviors of composite materials, with interface cracking being a typical failure event. The ability to precisely describe the mechanical behavior of the interface is critical for analyzing the failure of composite materials. Based on plasticity theory and damage mechanics, an elastic-plastic interface constitutive model is developed to simulate the irreversible plastic deformation at the interface under the cyclic loading condition. Furthermore, by incorporating a damage factor in scalar form, the model is able to simulate mixed loading forms (Modes I and II). The influence of interfacial strength and toughness on the overall mechanical properties of unidirectional fiber reinforced composites under transverse loading is studied with this model. The simulation results are consistent with the experimental data in the literature, demonstrating the model's effectiveness. The model can be used to simulate the mechanical behavior of composites with interfaces under cyclic loading, and it is helpful for understanding the plastic behavior and damage accumulation at the interfaces in the composites.

interface, cohesive zone model, plasticity, damage, fiber reinforced composites

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1 Introduction

The interfacial properties between fibers and matrix, including stiffness, strength and toughness, play important roles in the overall mechanical properties of composites [1–5]. The interfaces transmit the load from the matrix to the fiber, acting as a bridge between them. However, the interface with appropriate mechanical properties can prevent crack propagation and release the stress concentration. By modulating the interfacial properties, the overall mechanical properties of composites can be effectively enhanced. The fiber-matrix interface failure is one of typical failure modes in the fiber reinforced composites [6]. As a result, predicting the overall mechanical behaviors of the composite requires a

deep understanding of the mechanical behavior of the interface.

To avoid the singularity of crack tip in linear elastic fracture mechanics, the concept of cohesive zone model (CZM) is widely utilized to characterize the interface behaviors. For the classical CZM, the constitutive equation relates the traction to the crack opening displacement (COD) at the interface where a crack occurs, which can predict delamination initiation and propagation based on the strength analysis and the fracture mechanics, respectively. Researchers have proposed various types of interface constitutive laws, such as bilinear, trapezoidal, and exponential models. These different forms of constitutive law are widely used to describe the interfacial behavior in various composites. The CZMs proposed by Barenblatt [7] and Dugdale [8] consider different loading scenarios and provide corresponding con-

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stitutive relations for the interface bonding element. CZM combined with finite element method (FEM) is supposed to be an effective method for simulation of interface failure and delamination propagation of composite materials. A four-segment linear cohesive law based on the fiber bridging mechanism was presented by Yin et al. [9], which takes into account the influence of mixed-mode ratios and fiber bridging phenomena on fracture toughness. Lu et al. [10] developed an elastic-viscoplastic CZM based on the standard continuum description to provide a constitutive framework that considers viscoplastic deformation accompanied by isotropic hardening behavior at finite deformations. Xu et al. [11] proposed a CZM with explicit consideration of the post-debonding friction, which is implemented to investigate the shear deformation and frictional mechanisms for the fiber push-out tests. Sun et al. [12] proposed two rate-dependent CZMs, which can characterize viscoelastic behaviors within the fracture process zone ahead of the crack tip, to analyze interfacial fracture of composite sandwich beams.

However, the experimental results indicate that there is visible plastic deformation at the interface crack tip in the composites [13–15]. A few cohesive constitutive models that take into account interfacial plastic damage have been presented by researchers. In order to better capture the effects of mixed loading and unloading processes, Kolluri et al. [16] extended the classical CZMs by introducing two plastic parameters, and their simulation results are in good agreement with the experimental observation of adhesive interface delamination. Meng et al. [17] introduced a generalized damage factor into the bilinear CZM, allowing the model to be applied to a wide range of composite materials. However, changing the mixed loading ratio throughout the multiple loading and unloading process may result in events that do not follow the thermodynamic principle, such as the restoration of the cohesive state [18].

Until now, corresponding fracture toughness measurement standards have been developed to obtain the accurate parameters for the interface constitutive model, under opening, in-plane shearing, tearing, and combination loading conditions [19–21]. To consider the effect of mixed mode loading, researchers have developed a variety of mixed-mode damage evolution law, such as power law and B-K law [22], which have been embedded in commercial FEM software. To improve the accuracy of the simulation of the interface failure and interfacial crack propagation, it is of great significance to develop a cohesive constitutive model that takes into account the plasticity and damage of the interface, as well as the effects of mixed loading conditions.

In this paper, we establish an elastic-plastic CZM that takes plasticity and damage into account, which can be applied to the situations with the damage accumulation at interface under cyclic loading. At the same time, a scalar damage variable is introduced to simulate interface damage under

mixed-mode loading. The aforementioned CZM is also implemented into a commercial software package ABAQUS as a user-defined element (Section 2.1). The effectiveness of the presented CZM for mixed loading forms is evaluated by comparing it to bilinear CZM and B-K law (Section 2.2). Besides, it can also capture the irreversible unloading response of the interface (Section 2.3). A 2D FEM model of unidirectional fiber reinforced polymer (FRP) under cyclic loading is established, to reveal the effect of interfacial plastic displacement on the mechanical response of the composites. Finally, the influences of interfacial parameters on the overall mechanical properties of composites are discussed in detail (Section 3).

2 Models and methods

2.1 Incremental cohesive constitutive model and numerical realization

It is assumed that the traction stress is proportional to the elastic displacement, which is independent of the plastic displacement, and the interface opening displacement can be divided into elastic and plastic parts based on the plasticity theory [23]:

$$\delta = \delta_e + \delta_p, \delta_e = [\delta_n^e, \delta_s^e, \delta_t^e]^T, \delta_p = [\delta_n^p, \delta_s^p, \delta_t^p]^T, \quad (1)$$

where δ_e and δ_p are the elastic and plastic displacement column vector, respectively; n represents the normal direction, and s and t indicate the shearing and tearing directions, respectively.

$$\sigma = \varepsilon(D) \mathbf{K} \delta_e, \quad (2)$$

$$\varepsilon(D) = 1 - D, \quad (3)$$

where D is the damage variable, which has a value between 0 and 1; $\mathbf{K}$ is the stiffness matrix, which is a diagonal matrix if we ignore the coupling of stiffness in different directions. The interface stress is proportional to the opening displacement before the yield condition ($D=0$). The following prerequisites should be met to prevent the mutual penetration of crack surfaces:

$$\sigma_n = K_n \delta_n, \text{ if } \delta_n < 0, \quad (4)$$

where K_n is the normal stiffness. According to eq. (4), the normal compressive stress has no contribution to the interface damage.

In order to thoroughly assess the effects of the plastic displacement in opening and shearing directions, a scalar parameter is adopted to describe damage degree. Based on the assumption of scalar form of damage variables, the coupled damage criteria such as B-K criterion can be realized, with the damage variable defined as

$$D = 1 - e^{-\delta_{ep}}, \quad (5)$$

$$\delta_{ep} = \left(\frac{\delta_n^p}{\gamma_n} \right)^2 + \left(\frac{\delta_s^p}{\gamma_s} \right)^2 + \left(\frac{\delta_t^p}{\gamma_t} \right)^2, \quad (6)$$

where δ_{ep} is the equivalent plastic opening displacement; δ_n^p is the normal plastic displacement; δ_s^p and δ_t^p are the tangential plastic displacements; γ_n is the normal softening coefficient, representing the effect of the normal plastic opening displacement on the damage degree; γ_s and γ_t are the shear softening coefficients.

Referring to the Hashin-Rotem failure criterion [24], Ye et al. [25] proposed a quadratic failure criterion to describe the delamination damage of initial yield surface:

$$f(\sigma) = \left(\frac{\langle \sigma_n \rangle}{\sigma_n^0} \right)^2 + \left(\frac{\sigma_s}{\sigma_s^0} \right)^2 + \left(\frac{\sigma_t}{\sigma_t^0} \right)^2 - 1 = 0, \quad (7)$$

where σ_i^0 ($i = n, s, t$) are the failure stress in different directions. $\langle \cdot \rangle$ is the Macauley brackets, which is defined as $x = (x + |x|) / 2$. Based on the quadratic failure criterion and combined with the B-K criterion, Turon et al. [18] proposed an initial failure criterion taking into account the mixed loading conditions:

$$\begin{aligned} (\sigma^0)^2 &= (\sigma_n)^2 + (\sigma_s)^2 + (\sigma_t)^2 \\ &= (\sigma_n^0)^2 + \left((\tau_{\text{shear}}^0)^2 - (\sigma_n^0)^2 \right) B^\eta, \end{aligned} \quad (8)$$

where η is the semi-empirical exponential parameter of B-K criterion, and $\tau_{\text{shear}} = \sqrt{\sigma_s^2 + \sigma_t^2}$. The stress space, as illustrated in Figure 1, reflects the comparison between the maximum traction criterion and the previous two criteria.

It is shown that the initial failure surface is not seriously influenced by the mixed loading conditions. According to recent investigations [26,27], the quadratic failure criterion put out by Ye et al. [25] has a good performance in predicting

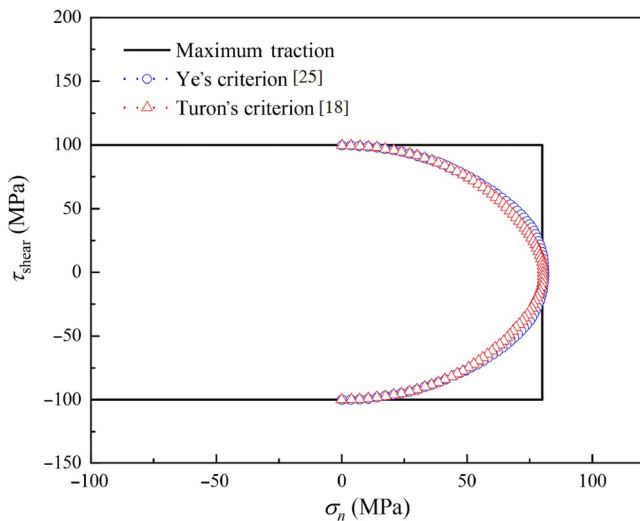


Figure 1 (Color online) Comparison between failure surfaces with different criteria.

the initiation of interface damage in FRP. Hence, the yield function can be expressed as

$$f(\sigma, \varepsilon(D)) = \left(\frac{\langle \sigma_n \rangle}{\sigma_n^0} \right)^2 + \left(\frac{\sigma_s}{\sigma_s^0} \right)^2 + \left(\frac{\sigma_t}{\sigma_t^0} \right)^2 - \varepsilon(D) = 0. \quad (9)$$

Finally, the direction of the plastic displacement vector can be obtained according to the plastic potential theory, using the orthogonal flow rule:

$$d\delta_p = d\lambda \frac{\partial f}{\partial \sigma}, \quad (10)$$

where $d\lambda$ is the plastic flow factor. Differentiating eq. (2), we can obtain

$$\begin{aligned} d\sigma &= \varepsilon(D) \mathbf{K} d\delta_e + \mathbf{K} \delta_e d\varepsilon(D) \\ &= \varepsilon(D) \mathbf{K} (d\delta - d\delta_p) + \mathbf{K} \delta_e d\varepsilon(D). \end{aligned} \quad (11)$$

The first-order Taylor expansion of $f(\sigma, \varepsilon(D))$ at point $(\sigma_a, \varepsilon(D)_a)$ can be expressed as

$$(d\sigma)^T \frac{\partial f}{\partial \sigma} \bigg|_{f(\sigma_a, \varepsilon(D)_a)} + d\varepsilon(D) \frac{\partial f}{\partial \varepsilon(D)} \bigg|_{f(\sigma_a, \varepsilon(D)_a)} = 0. \quad (12)$$

The consistency condition is determined as eq. (12). Combining eqs. (10)–(12), we can get

$$\begin{aligned} d\lambda &= \frac{\left(\frac{\partial f}{\partial \sigma} \right)^T \mathbf{K} \varepsilon(D) d\delta}{\left[\left(\frac{\partial f}{\partial \sigma} \right)^T \mathbf{K} \varepsilon(D) - \left(\frac{\partial f}{\partial \varepsilon(D)} + \left(\frac{\partial f}{\partial \sigma} \right)^T \mathbf{K} \delta_e \right) \left(\frac{d\varepsilon(D)}{d\delta_p} \right)^T \right] \frac{\partial f}{\partial \sigma}}. \end{aligned} \quad (13)$$

The incremental constitutive relation for the plastic phase can be obtained as

$$\begin{aligned} &\left[\left(\frac{\partial f}{\partial \varepsilon(D)} + \left(\frac{\partial f}{\partial \sigma} \right)^T \mathbf{K} \delta_e \right) \left(\frac{d\varepsilon(D)}{d\delta_p} \right)^T - \left(\frac{\partial f}{\partial \sigma} \right)^T \mathbf{K} \varepsilon(D) \right] d\sigma \\ &= \frac{\partial f}{\partial \varepsilon(D)} \left(\frac{d\varepsilon(D)}{d\delta_p} \right)^T \mathbf{K} \varepsilon(D) d\delta. \end{aligned} \quad (14)$$

Based on the aforementioned theory, the elastic-plastic interface constitutive model is realized using the user sub-routines in the FEM software ABAQUS by following the procedures in Figure 2. A four-node two-dimensional interface element (COH2D4) from the ABAQUS explicit module is adopted in the following simulations. Figure 3(a) depicts the stress-displacement curve under the normal tensile loading and unloading, with the parameters listed in the Table 1. As illustrated in Figure 3, the incremental CZM accurately describes the process of continuous damage and entire failure of the interface, with obvious nonlinear characteristics. With irreversible plastic deformation, interface degradation causes a nonlinear decrease in stress and stiffness. To examine the mechanical behavior of CZM established in this paper under cyclic loading, the cyclic shear load is applied to the model. As shown in Figure 3(b), it is clear

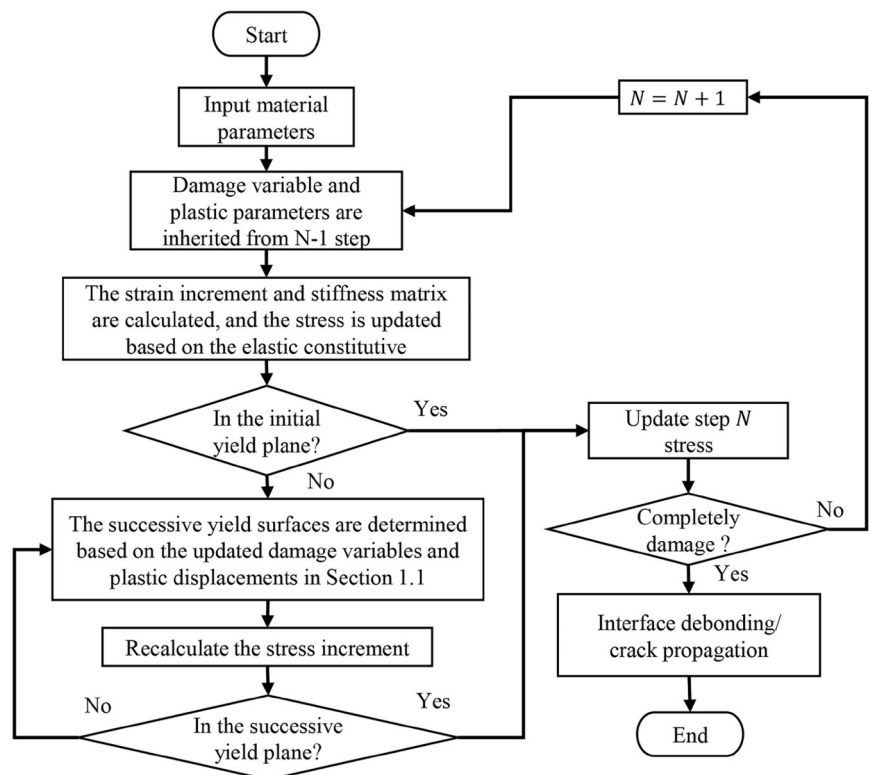


Figure 2 Implementation process of constitutive model.

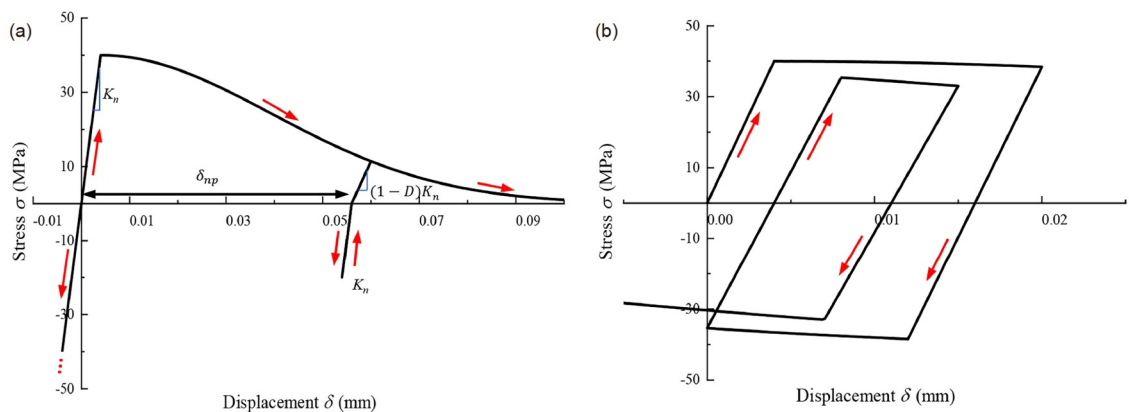


Figure 3 (Color online) Traction-separation curve under cycle loading. (a) Mode I; (b) Mode II.

Table 1 Interface constitutive parameters

CZM	K_n (MPa mm ⁻¹)	$K_s=K_t$ (MPa mm ⁻¹)	σ_n^0 (MPa)	$\sigma_s^0 = \sigma_t^0$ (MPa)	γ_n (mm)	$\gamma_s = \gamma_t$ (mm)
	1×10^4	1×10^4	40	40	0.05	0.08

that the interface shows a strain-softening effect.

2.2 Mixed-mode loading

Experimental techniques were applied to access the fracture

toughness of several types of adhesive layers under various mixed-mode ratios, and a corresponding cohesive model was developed based on the J-integral methodology [15]. Here, we establish a finite element model in ABAQUS with the COH2D4 element using the incremental CZM presented in

this paper. The fracture toughness under various mixed-mode ratios is then obtained by numerical simulations, as shown in Figure 4, after setting the consistent fracture toughness with experimental results [15] under the single loading mode. The mixed-mode ratio is defined as [22]

$$\lambda = \frac{G_I}{G_T}, \quad (15)$$

where G_I and G_T are the strain energy release rate for the Mode I and total, respectively. As a comparison, fracture toughness under different mixed-mode ratios conditions could be also obtained using the bilinear CZM combined with B-K law. According to Figure 4, it is obvious that the incremental CZM developed in this study is capable of accurately capturing the effects of mixed loading conditions.

2.3 Plastic damage of interface under unloading

The prior research provided ample evidence that plasticity frequently occurs in adherent layers, especially if the layers comprising the interface are ductile and the interface is strong [28,29]. A new bilinear CZM proposed by Meng et al. [17] considered not only the coupling of negative normal traction and tangential traction, but also plastic damage during loading and unloading by changing the generalized damage factor. An irreversible plasticity-damage model was introduced by Kolluri et al. [16] and can be used to augment the current CZMs with an unloading description that is appropriate for modeling the unloading response. Accordingly, the incremental elastic-plastic CZM developed in this study takes interface damage into account and may be utilized to simulate mixed-mode loading conditions, exhibiting excellent computational efficiency.

Kolluri et al. [16] extended an improved Xu-Needleman CZM [30] with the combined plasticity-damage formulation, which has been validated by experiments. Numerical simulations with this extended model are performed for the copper lead frame-molding compound epoxy (CuLF-MCE) interface for comparison, with the sample geometry and loading conditions illustrated in Figure 5. The material properties of the adherent structural layers and the interface (Mode I) samples are listed in Tables 2 and 3, which are also

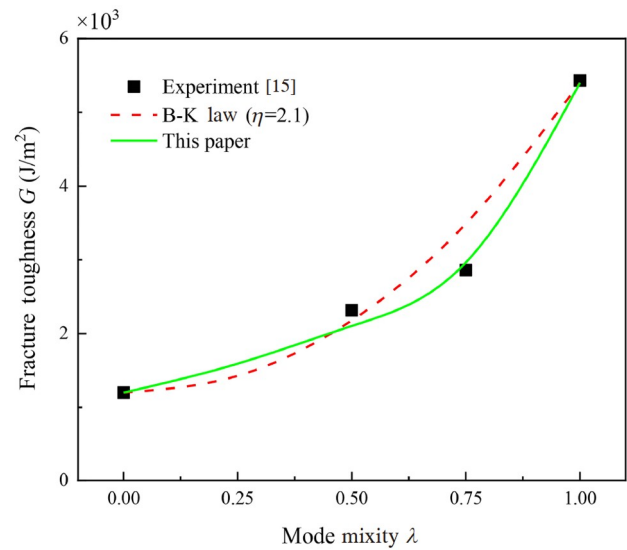


Figure 4 (Color online) Variation of fracture toughness under mixed loading conditions.

Table 2 Material properties of CuLF and MCE [31]

Material parameters	CuLF	MCE
Young's modulus (GPa)	120	30
Poisson's ratio	0.33	0.25

adopted from ref. [31]. The traction-separation curves of the three interface constitutive models (in Mode I) are shown in Figure 6. It can be seen that the CZM extended by Kolluri [31] has the exponential traction-separation law with a nonlinear damage evolution stage, which is similar to the CZM proposed in this paper. While, the bilinear CZM has a linear damage evolution stage on the contrary.

In the following numerical investigation, the ABAQUS/Standard is employed, and the CuLF and MCE are assigned with the CPE4R reduced integral plane strain element. The COH2D4 elements with zero thickness are then introduced between them to simulate the interface debonding process. With mesh convergence verification, approximately 10000 elements with a mesh size of 0.1 mm are adopted in the numerical simulations. Finally, Figure 7 displays the load

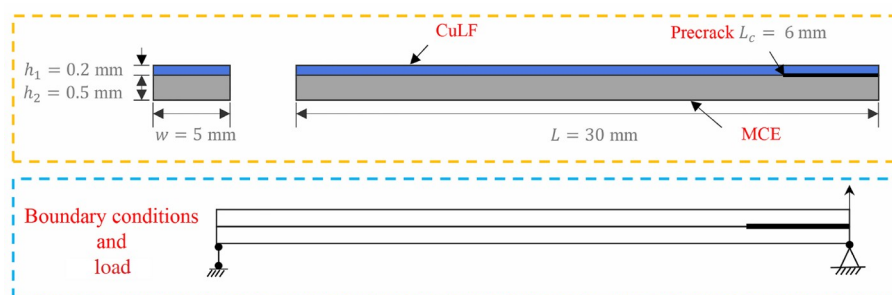


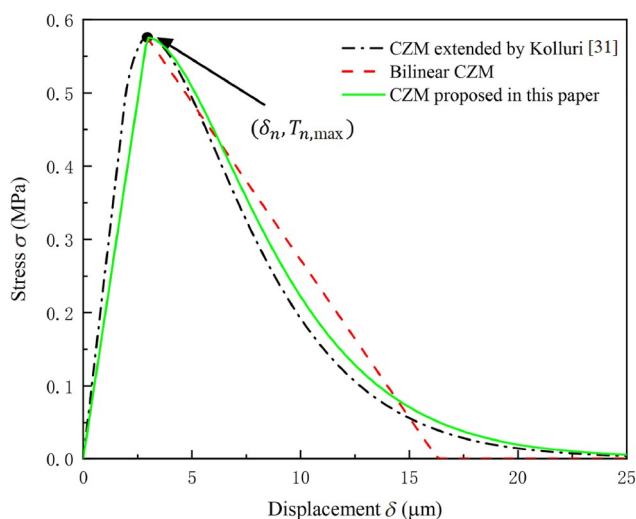
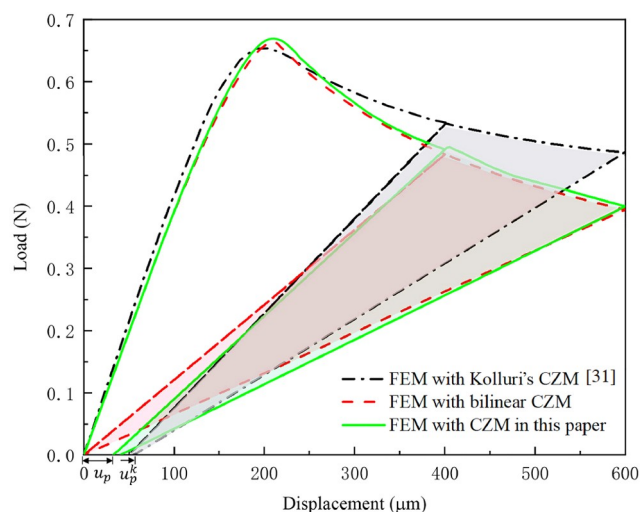
Figure 5 (Color online) Geometry and the applied boundary conditions of the CuLF-MCE model.

Table 3 Material properties of interface^{a)}

Kolluri's CZM [31]	$T_{n,max}$ (MPa)	δ_n (μm)	G_I (J/m^2)	δ_{pl} (μm)
	0.576	3	4.7	10
Bilinear CZM	K_n (MPa mm^{-1})	σ_n^0 (MPa)	G_I (J/m^2)	–
	196	0.576	4.7	–
Our CZM	K_n (MPa mm^{-1})	σ_n^0 (MPa)	γ_n (mm)	–
	196	0.576	0.0038	–

a) δ_n and $T_{n,max}$ are the parameters of the Xu-Needleman CZM. G_I is the critical energy release rate in Mode I. δ_{pl} is a parameter of the Kolluri's extended model [16].

displacement responses obtained using the three different CZMs. When Kolluri's or our CZM is employed to describe

**Figure 6** (Color online) Comparison between the normal traction-separation law of three CZMs.**Figure 7** (Color online) Loading-unloading load-displacement responses of CuLF-MCE interface obtained by different CZMs under Mode I delamination.

the interface, it can be observed that there are substantially retained openings indicating the presence of plasticity and damage at the interface, which is consistent with the experimental observation [29]. However, the simulation based on the bilinear CZM cannot produce comparable results. By setting the unloading displacement at 0.4 and 0.6 mm, respectively, we can determine the total energy lost between two successive unloading processes. It also can be seen that the retained opening value based on Kolluri's CZM (u_p^k) is slightly greater than that based on presented model (u_p). One possible reason is that Kolluri [31] considered the contribution of structural plasticity in their simulations.

3 Results and discussion

Unidirectional FRP under transverse tensile loading is investigated for the progressive damage evolution, using the incremental interface constitutive model developed in Section 2.1. Based on the random collision algorithm [32], a representative volume element (RVE) with 50 fibers [33] and 60% fiber volume fraction is constructed. For generation of the RVE model, the fiber diameter is set as 16 μm , and the side length of square RVE is 129 μm . Besides, the interface thickness is 0.1 μm , which is far less than the diameter of fiber. For the following numerical study, the ABAQUS/explicit integral model is employed to avoid convergence problems. CPE4R reduced integral plane strain elements are assigned for the fiber and matrix, and COH2D4 interface elements are inserted between them to simulate the interface debonding process.

Taking into account the collision model's randomness, 10 distinct RVE finite element models, each with approximately 25000 elements, are generated. Figure 8 depicts one of these models. Periodic boundary conditions (PBC) are adopted on the model's boundary as [32]

$$\mathbf{U}_R - \mathbf{U}_L = \mathbf{U}_{RB} - \mathbf{U}_{LB}, \quad (16)$$

$$\mathbf{U}_T - \mathbf{U}_B = \mathbf{U}_{LT} - \mathbf{U}_{LB}, \quad (17)$$

where $\mathbf{U}$ is the displacement vector of the node on the

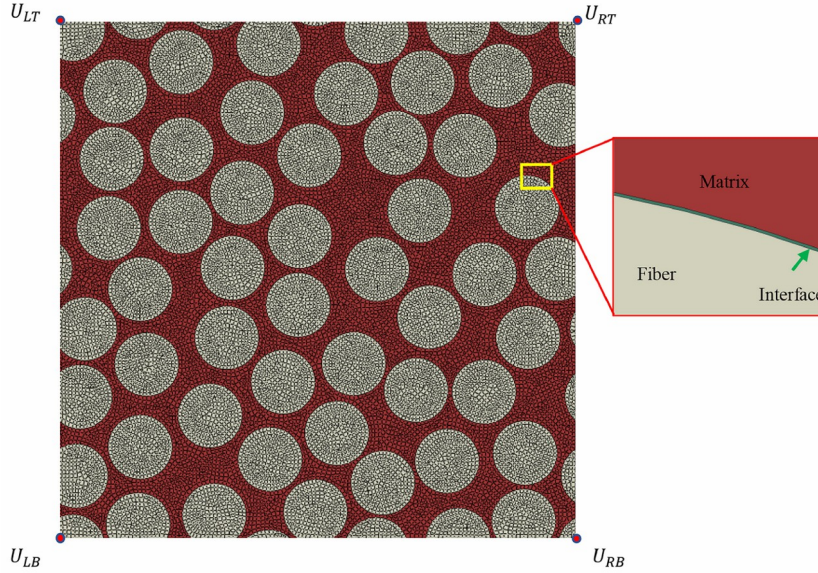


Figure 8 (Color online) RVE geometry model with periodic boundary conditions.

boundary. Right, left, up, and down directions are denoted by R , L , T , and B , respectively. Specific corner nodes are indicated by double subscripts, as shown in Figure 8.

3.1 model parameters

Matrix damage and fiber-matrix interface failure are the two main damage modes of FRP when subjected to transverse loads. This paper focused on the glass fiber reinforced epoxy resin composites (E-glass/MY750), in which fibers are regarded as isotropic materials in the plane strain model, with a Young's modulus (E_f) of 74 GPa and a Poisson's ratio (ν_f) of 0.2 [34]. The linear Drucker Prager model [2] is adopted to describe the plastic behavior of polymer matrix, since experiments demonstrate that hydrostatic pressure has a significant effect on the yield behavior of polymer materials [35], and the tensile and compressive yield stresses are usually not the same.

$$F = t - p \tan \beta - d = 0, \quad (18)$$

where p is hydrostatic pressure; β is the slope of the yield surface in p - t stress plane, commonly known as friction angle of the material; d is the cohesion strength; t can be expressed as

$$t = \frac{1}{2}q \left[1 + \frac{1}{\Gamma} - \left(1 - \frac{1}{\Gamma} \right) \left(\frac{r}{q} \right)^3 \right], \quad (19)$$

where r is the third invariant of the deviatoric stress; q is the von Mises equivalent stress; Γ is the ratio of yield stress in triaxial tension to triaxial compression.

The linear Drucker Prager model is formulated in terms of all three stress invariants, which provides a possibly non-circular yield surface in the deviatoric plane to match dif-

ferent yield values in triaxial tension and compression. It is useful when the experimental data have already been calibrated in terms of cohesion and friction angle. When these parameters are supplied for a Mohr-Coulomb model, they must be converted to Drucker-Prager parameters.

Previous studies [2,4,5] have shown that the input material properties for the Drucker-Prager plasticity model can be calculated using the relation between the Mohr-Coulomb and the Drucker-Prager material constants. According to the Mohr-Coulomb plasticity criterion, the cohesion stress and the angle of internal friction are related to the tensile and compressive strength of the material by [5]

$$\sigma_m = 2c \frac{\cos \phi}{1 + \sin \phi}, \quad \sigma_t = 2c \frac{\cos \phi}{1 - \sin \phi}. \quad (20)$$

The relevant values are listed in Table 4, where E_m and ν_m are the Young's modulus and Poisson's ratio of the matrix, respectively; ϕ is the angle of internal friction; c is the cohesion stress; σ_t and σ_c are the yield stress of tension and compression, respectively. The equation of the Mohr-Coulomb model can be used to calculate the parameters of the linearly extended Drucker Prager model under the assumption of plane strain [2].

$$\tan \beta = \frac{3 \sin \phi}{\sqrt{3 + \sin^2 \phi}}, \quad \Gamma = \frac{3 - \sin \phi}{3 + \sin \phi}. \quad (21)$$

According to the above formula, $\beta=18.98^\circ$, $\Gamma=0.875$ can be obtained. The material is considered as an ideal elastoplastic material, and the equivalent plastic strains for damage initiation are set to 0.05 and 0.5 [2] in uniaxial tension and compression cases, respectively. According to the experiments reported in refs. [36,37], the fracture energy of the matrix is taken as $G=1 \text{ J/m}^2$. According to the conclusions

obtained by Sharma et al. [4], the strength of materials under uniaxial tensile load is primarily determined by the interface’s fracture strength and fracture toughness under normal tensile loading. The mechanical parameters of interface under shear loading have little effect on the simulation result. Therefore, the cohesive constitutive parameters are listed in Table 4 [4,34,38]. Python is used to implement the program that calculates the parameters of the constitutive model to determine the interface fracture toughness. It is verified that the fracture toughness obtained by taking the interface constitutive parameters in Table 4 is consistent with the optimized fracture toughness in ref. [4], as $G_I = G_{II} = 2 \text{ N/m}$.

3.2 Simulation results

3.2.1 Uniaxial tensile loading

As indicated in Figure 9, lateral displacement load is applied

to the two-dimensional model. Because the interface’s strength is lower than the matrix’s yield strength, when the average tensile stress exceeds roughly 40 MPa, damage initiates at the local fiber matrix interface and cracks propagate along the interface, continuing to expand along the arc-shaped interface. When the interface cracks extend to a debonding angle (ψ) of about 60°–70°, the cracks propagate in an unstable manner. Finally, the micro cracks along the interface have a propensity to develop into a main crack, which eventually leads to the overall collapse of the composite material. This is basically consistent with the experimental phenomenon observed by Paris et al. [39] using scanning electron microscopy (SEM). Based on the FEM simulations with 10 distinct models, the debonding angles for kinking are most logically between 60° to 70°, which is confirmed by both numerical simulations and experiments.

Figure 10 shows a comparison between the predicted

Table 4 Matrix material Mohr-Coulomb model parameters [34,38] and interface constitutive parameters [4]

Matrix	E_m (GPa)	ν_m	ϕ (°)	c (MPa)	σ_t (MPa)	σ_c (MPa)
	3.35	0.35	11.53	49	80	120
CZM	K_n (MPa mm ⁻¹)	$K_s=K_t$ (MPa mm ⁻¹)	σ_n^0 (MPa)	$\sigma_s^0 = \sigma_t^0$ (MPa)	γ_n (mm)	$\gamma_s = \gamma_t$ (mm)
	1×10^8	1×10^8	53	53	0.05	0.05

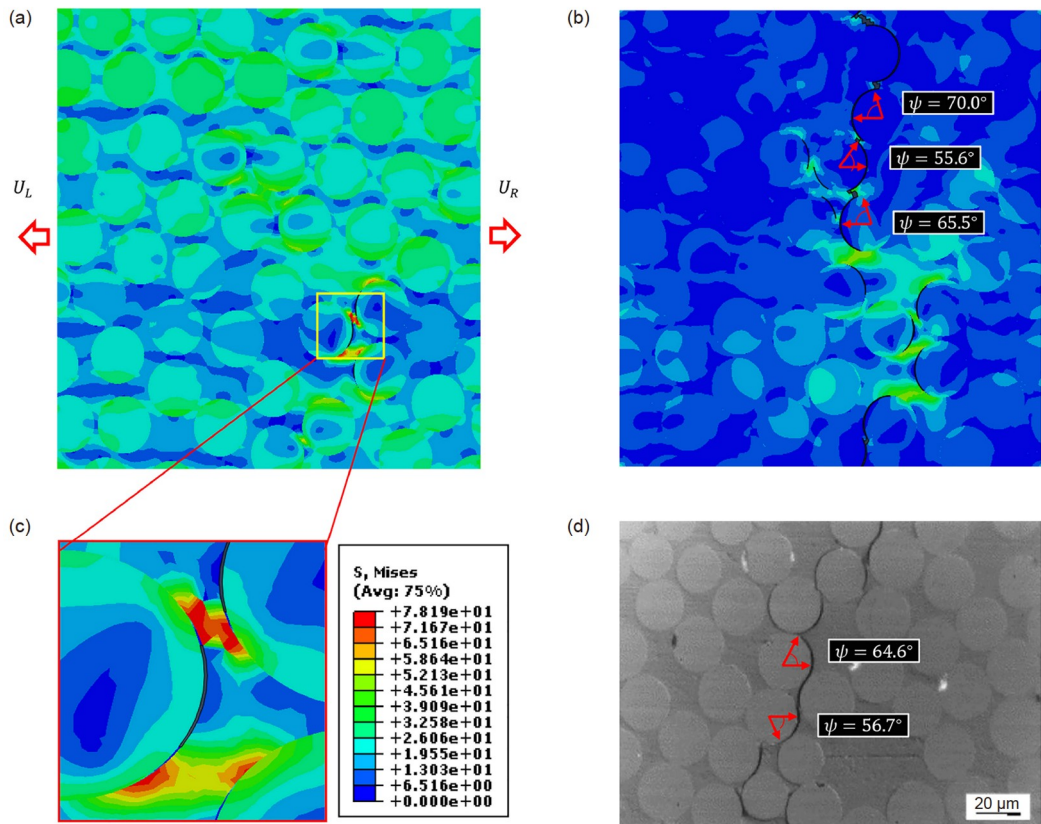


Figure 9 (Color online) (a) Initial damage at the interface; (b) crack propagation along the interfaces; (c) close view of damage; (d) experimental observation [39].

average stress-strain curve and the results of earlier simulations and experiments for unidirectional composites subjected to transverse uniaxial tensile loading. We establish a set of stochastic finite element models to take into consideration the randomness of the models. The obtained average stress-strain curve is illustrated by the black solid line in Figure 10. The results of Sharma's simulations [4] based on the bilinear CZM and experimental curves [40] are compared with the calculation results based on this model. The good agreement between the simulation and experiment validates the incremental CZM proposed in this study.

3.2.2 Cyclic loading and unloading

In contrast to the traditional CZMs [7,8], the incremental cohesive constitutive model proposed in this study can be adopted to simulate the plastic damage of the fiber matrix interface during cyclic loading and unloading. Figure 11 illustrates the stress-strain curve under uniaxial tensile loading and unloading conditions with interface strength of 40 MPa (less than the matrix tensile strength 80 MPa) and fracture toughness of 5 J/m^2 . For the bilinear CZM, the stress and strain can always return to the initial state (origin point) when unloading, even if the interface has entered the damage state, as shown in Figure 11.

However, as soon as the interface reaches the plastic status, there will unavoidably be residual deformation, which is actually irreversible. The incremental CZM suggested in this work is capable of simulating this phenomenon with high accuracy, and the damage mechanism can be reasonably revealed. In the loading process, with the increase of local stress level, the local fiber matrix interface firstly reaches plasticity stage, with the damage beginning. Residual deformation is caused by interface plasticity, and clear stress concentration can be observed in a local matrix, which is comparable with the behavior described in refs. [4,41].

The stress distributions in the composites for the highlighted spots A and B in Figure 11 are shown in Figure 12(b) and (e), respectively. Point A is in the unloading process based on the bilinear cohesive model, while point B is in the unloading process based on the CZM presented in this paper, and they are both in the same strain state. Obviously, Figure 12(e) shows a more significant stress concentration when the interface is under unrecoverable plastic deformation due to the stress redistribution caused by local plasticity.

3.2.3 Influence of the interface parameters

The normal tensile strength and fracture toughness of the interface are evaluated as essential characteristics to assess the impact of interface parameters on the overall mechanical properties of composites. Firstly, by setting the fracture toughness as 5 J/m^2 , the composite mechanical properties can be obtained under the different fiber-matrix interface strengths, as shown in Figure 13. With the increase of in-

terface strength, and from weak interface change to strong interface, the composite material shows a transition from elastic-plastic to elastic-brittle behavior. The initial elastic phases of the stress-strain curves of composites with different interfacial strengths almost coincide, as shown in Figure 13(c). The transverse tensile stress-strain curve of the composite exhibits a clear plastic stage if the interface strength is lower than the matrix's tensile strength, as seen in Figure 13(a). On the contrary, a portion of the matrix will initially enter the plastic stage if the interface strength is larger than the tensile strength of the matrix, as shown in Figure 13(b). The composites as a whole exhibit an apparent brittle failure mode, which is consistent with the conclusion reported by Tan et al. [1].

The composite material may quickly lose its bearing cap-

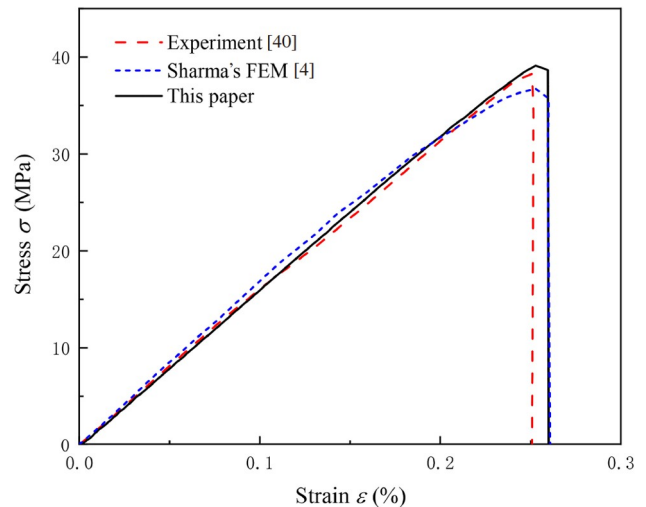


Figure 10 (Color online) Comparison of the lateral tensile stress-strain curves.

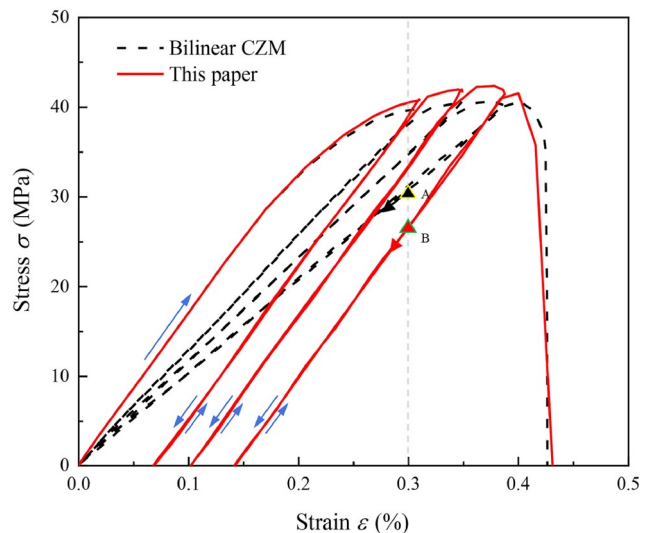


Figure 11 (Color online) The stress-strain curves under cyclic loading and unloading.

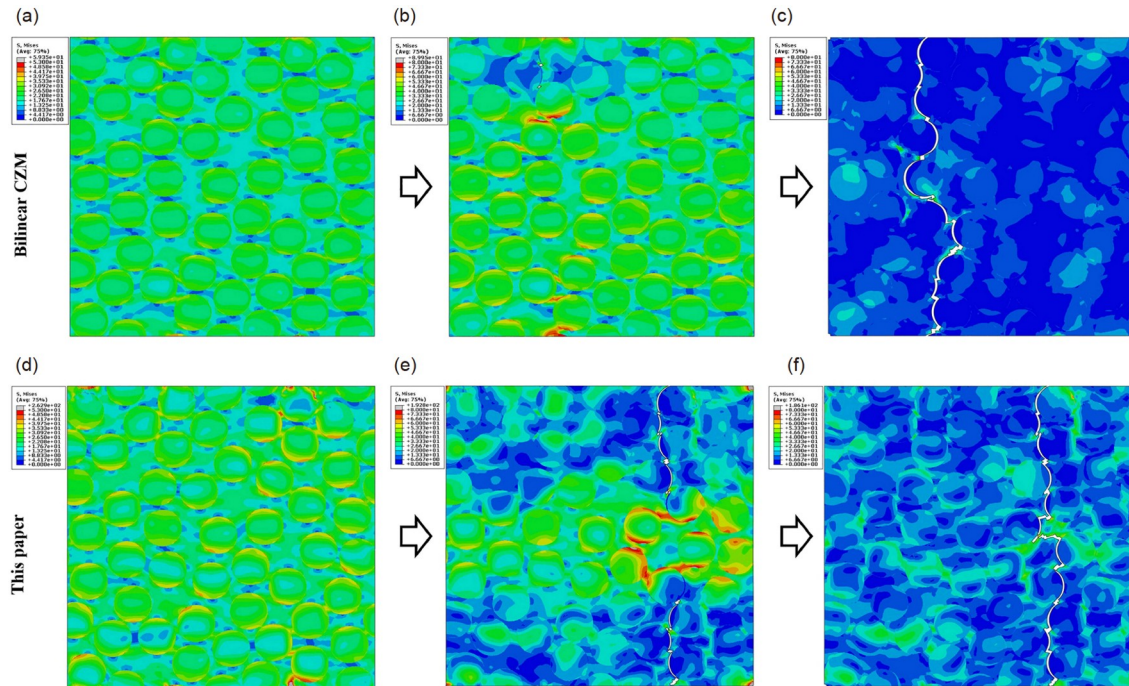


Figure 12 (Color online) Based on the bilinear CZM, the stress distribution at elastic deformation (a), interface debonding at (b) and then propagating through the specimen (c). With the incremental CZM presented in this paper, the stress concentration is obvious at the interfaces (d), and the crack nucleates at interface (e) and then propagates along the bridged interfaces, with residual stress left (f).

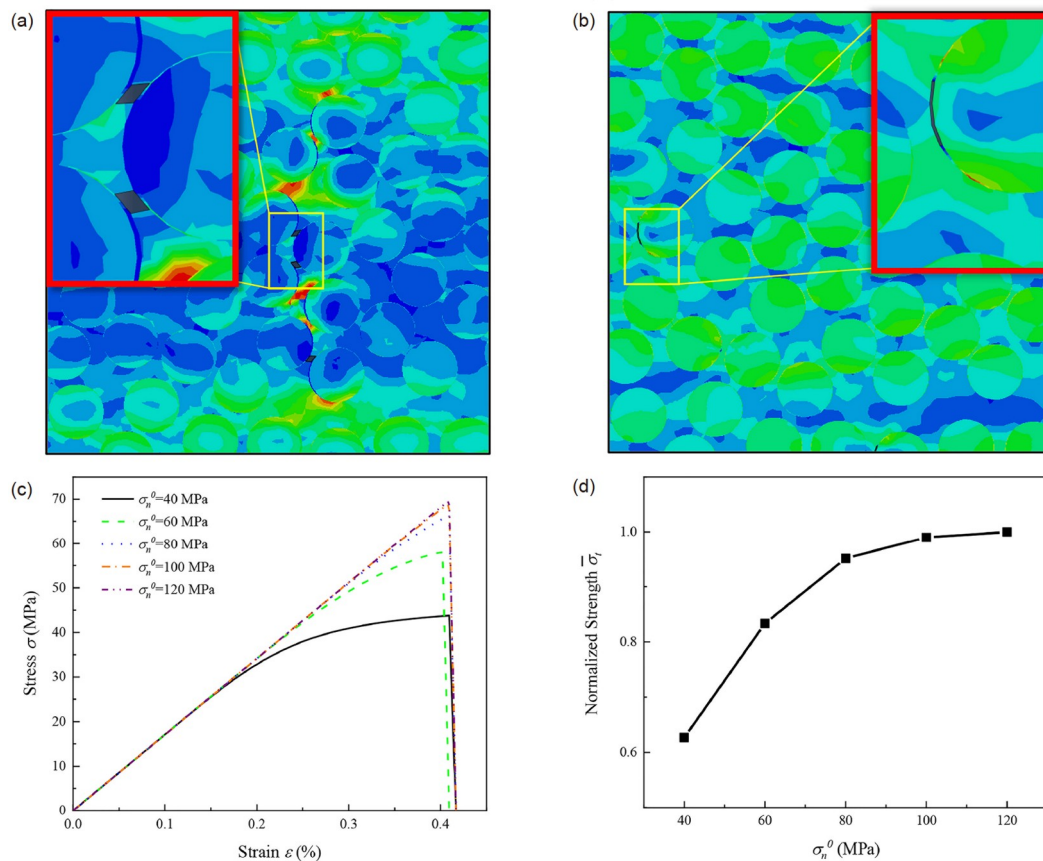


Figure 13 (Color online) (a), (b) Initial damage stress nephogram with interface strength of 40 and 80 MPa, respectively. (c) The stress-strain curves of single cell with different interface strengths, and (d) the effect of interfacial strength on the normalized strength of composites.

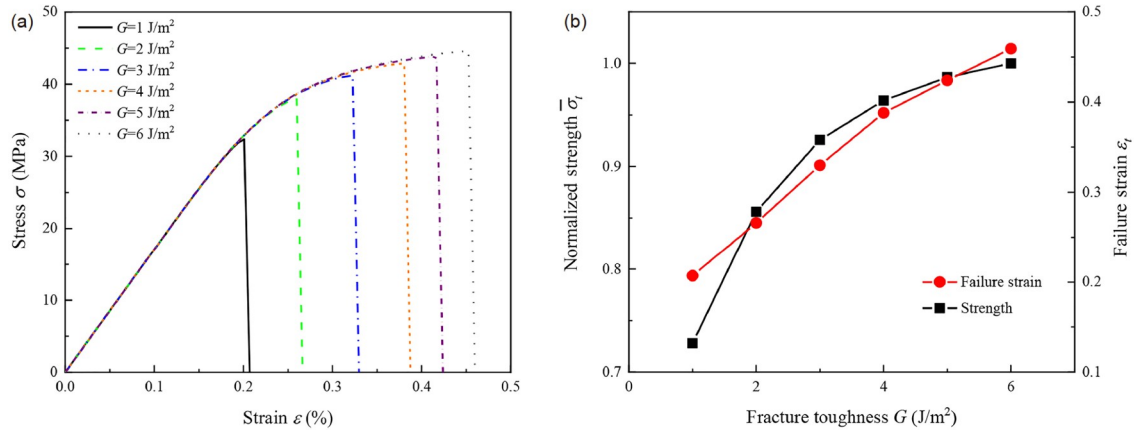


Figure 14 (Color online) (a) The stress-strain curves with different fracture toughness; (b) the effect of fracture toughness of interface on the normalized strength and failure strain of composites.

ability if the matrix fails. Figure 13(d) illustrates the correlation between the composite strength and the interface strength obtained by finite element simulation. The strength of composites is normalized by the maximum strength ($\sigma_{t_{\max}}$). As can be seen, when the interfacial strength is lower than the matrix's yield strength, the composite's uniaxial tensile strength can be enhanced significantly by increasing the interfacial strength. While altering the interfacial strength has little impact on the composite's strength when the interfacial strength is higher than that of the matrix.

The transverse tensile stress-strain curves with different interfacial fracture toughness also can be obtained by setting the interfacial strength to 40 MPa, as illustrated in Figure 14(a). It is obvious that improving interfacial fracture toughness can enhance the strength of composites to a certain extent, since higher fracture toughness leads to higher failure strain, as shown in Figure 14(b).

4 Conclusion

In this study, an incremental CZM is established by integrating plastic theory and damage mechanics. The incremental interface constitutive model suggested in this study can be utilized to simulate the irreversible plastic displacement of the interface under cyclic loading, as opposed to the traditional bilinear CZM. Additionally, without incorporating other criteria like the B-K law, the new model put forward in this study can be applied to the interfaces under the mixed-mode loadings. With the consideration of interfacial phases, this new model is successfully used to simulate the damage evolution behavior of 2D fiber-reinforced composites under lateral stress. Finally, a thorough discussion of the impact of interface mechanical characteristics parameters on composites' overall mechanical properties is conducted.

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